

SPECIAL ISSUE ARTICLE



Telemedicine virtual reality based skin image in children's dermatology medical system

Juanjuan Gao | Chong Lyu | Xianhua Qiao | Fei Tian 

Dermatology Department, Xingtai
People's Hospital, Xingtai, China

Correspondence

Fei Tian, Dermatology Department,
Xingtai People's Hospital, Xingtai, Heibei
054001, China.

Email: tianfeiyisheng@163.com

Abstract

At present, with the rapid development of computer and communication technology and people's demand for medical technology services, telemedicine virtual reality technology has become a high-tech hot spot in the information age and a new trend in the development of the medical industry. However, my country's high-quality medical resources are mainly concentrated in large and medium-sized cities, and there is an urgent need to improve the fairness of medical services. Telemedicine virtual reality technology has received more and more attention from people, has been widely used worldwide, and has become an important part of hospital information engineering. This article first introduces the basic concepts of telemedicine virtual reality technology, the latest development of telemedicine virtual reality technology and related technologies, and then introduces the system in detail. The system is functionally divided into two parts: consulting management and system management. Consulting management includes all remote business consulting processes; system management includes management of medical resources and user data, and then detailed description and visualization of the development and use of disease model libraries. Telemedicine completely separates the rehabilitation area of skin patients from the hospital, and patients can complete relief and rehabilitation outdoors or even when they are discharged from the hospital.

As a remote access device, telemedicine virtual reality technology has been popularized to a certain extent. The general patient group represented by patients with stable skin conditions has improved their understanding of telemedicine technology, which is beneficial to the development of telemedicine virtual reality technology. The utilization rate, safety, rehabilitation effect and completion rate of telemedicine virtual reality technology are as high as 85%, 92%, and 96%, respectively, saving a lot of time and medical expenses. Therefore, telemedicine virtual reality technology is also the direction of my country's next medical reform, and it can also be used in other aspects of continuing education of telemedicine virtual reality technology.

KEYWORDS

children's skin diseases, remote dermatology, skin imaging, telemedicine technology

1 | INTRODUCTION

1.1 | Background and significance

With the rapid development of computer network technology and the advancement of medical technology, people pay more and more attention to medical health. In the medical field, the application of telemedicine virtual reality technology in the medical field has become a hot spot today. It uses various digital transmission methods through computer networks, multimedia technology, and telemedicine software systems to establish connections between medical units, physicians and patients in different fields, and complete remote consultation, diagnosis and treatment, teaching, academic research, and information.^{1,2} This work is a brand new model of diagnosis and treatment between medical experts and patients to help doctors make diagnoses faster and more accurately. Telemedicine consultation is a closer connection between medical experts and patients, so even if patients are far away, patients can more easily seek advice and treatment from hospital experts, which reduces the corresponding cost of patient visit time. At the same time, the remote consultation system also helps to learn medicine remotely. With the current increase in people's income levels, people's awareness of personal care has greatly improved, which to a certain extent also increases the demand for medical resources.³ Therefore, designing a telemedicine system has high practical significance and value.

1.2 | Related work research

Elsner evaluated his dermatological diagnosis and believed that there are few studies on its effectiveness, diagnostic accuracy, feasibility, and cost-effectiveness in dermatology. However, it



shows great potential for the diagnosis, prevention, treatment support, and follow-up of patients with skin diseases. Asynchronous or synchronous dermatology remote consultations cannot only help medical experts in prevention, but also help outdoor workers during skin cancer screening. Therefore, remote dermatology may help early prevention and notification of occupational skin diseases. Modern smartphone applications with artificial intelligence technology can also promote the self-monitoring of high-risk workers.⁴

Hsu believes that the analysis of skin quality through the telemedicine system has become an emerging trend. Digital images will be used to record facial skin, and image processing technology will be used to perform skin inspections to help doctors provide medical treatment to remote patients. After finding that the skin image was obtained, the roughness, freckles, and acne indicators were evaluated. These three indicators are used as the input of the system, and then the skin score is calculated to evaluate the skin condition to further provide more matching skin care. It can improve the health problems that have occurred, and can also record the skin condition of each test. The experimental results show that it is suitable for telemedicine applications.

1.3 | Related work research

1. This research further proves that it is more convenient to change the quality of life, exercise and adherence to treatment of patients with skin diseases in the Internet + era. Exploring the role of telemedicine technology in the treatment of children's skin diseases has greater practical significance for improving their prognosis and quality of life.
2. This research is characterized by low cost and high visibility. Use cloud ICU as a platform to provide face-to-face telemedicine technical support between physicians and patients. As a skin disease patient, you only need a computer or a smart phone to use it, thus avoiding diagnostic consequences such as infection.
3. This article is based on advanced communication technology and multimedia technology, and considers the balance between end-user input costs and technical means to achieve a real-time telemedicine system that can more effectively meet the needs of medical applications. With the continuous development of communication technology and multimedia technology and the continuous progress of society, the dream of telemedicine will soon be realized.

2 | APPLIED RESEARCH ON CHILDREN'S SKIN DISEASES MEDICAL SYSTEM

2.1 | Skin imaging

Skin imaging technologies currently mainly include reflective confocal microscope (RCM), dermoscopy, and skin ultrasound. RCM is called skin CT for short. It is a real-time and dynamic three-dimensional (3D) real-time skin imaging technology based on the principle of coherence. This is a noninvasive imaging test that can detect abnormal changes at the level of tissue cells. Dermatoscope is also called a microscope, which transmits light to the skin surface, has the characteristics of magnifying and eliminating the reflected light on the skin surface, and can observe the pathological changes of the skin at the submicroscopic level, such as the epidermal structure, pigment substances, blood vessels, and skin accessories.⁵



Evaluating the therapeutic effect of skin diseases is an important part of clinical work. Skin imaging technology can provide in vivo, noninvasive, real-time, and dynamic pathological changes of skin lesions. In order to assess skin damage before clinical removal of skin lesions, dermoscopy, RCM, and B-ultrasound can be used to provide relevant information, and dermoscopy can provide general information to observe whether there are any relevant changes in skin lesions. RCM can provide more detailed pathological changes, such as blood vessel structure. In clinical practice, most hemangioma is found, skin B-ultrasound can provide more detailed depth, and can determine the target during the operation. For postoperative detection of tumor diseases, it can effectively assess the resection status of skin lesions.^{6,7}

2.2 | Telemedicine technology

Telemedicine is also called telemedicine, it is the application of computer, communication, medical technology, and equipment, through long-distance data, text, voice, and image transmission, to realize real-time or nonreal-time between experts and patients, experts and medical staff. Exchange information in different places. Telemedicine is not only a medical or clinical issue, but also includes communication networks, databases, computer science, and other aspects, which must be integrated into network systems. In short, telemedicine is the product of the combination of modern information technology and modern medical technology.^{8,9} It usually includes remote diagnosis, remote consultation, remote education, and remote medical information exchange.

Telemedicine is the application of computer network technology in the medical field. With the rapid development of computer network technology, multimedia technology, and communication technology, telemedicine systems have been widely used. In recent years, on the basis of in-depth analysis of 3D medical images, medical imaging databases, and hospital HIS systems, a large number of related systems have been researched and developed, and many results have been achieved. Based on this background, this article focuses on system architecture, multimedia communication, real-time voice transmission, WEB-based electronic medical records, collective diagnosis, and other key technologies related to the design and implementation of telemedicine systems.¹⁰

Telemedicine is very important to improve the living standards of our people. Telemedicine includes distance diagnosis, expert consultation, intelligence service, online inspection, and distance education. High-quality telemedicine can help improve the health of everyone, especially the care of the elderly and patients with chronic diseases. At the same time, interregional consultations can be conducted on difficult cases to effectively improve the quality of medical care in certain areas. Telemedicine consultation has established a new connection between medical experts and patients, enabling patients to receive advice for treatment and care under the guidance of remote experts in primary hospitals and first-time hospitals, thus saving doctors and patients a lot of time and money.^{11,12}

2.3 | Application of telemedicine in dermatology

Teleradiology is one of the earliest and most widely used applications. Hospitals in remote areas send CT images, X-rays, and magnetic resonance imaging to hospitals in big cities, and consult relevant experts. Telepathology Through a controlled microscope connected to a remote



communication device, a remote doctor can make a quick and accurate diagnosis.¹³ This application does not require sending pathological samples to remote laboratories. Similar to radiology, dermatology is also very suitable for telemedicine applications. Because the diagnosis of skin conditions mainly depends on morphology, both clinical manifestations and pathological incisions can be processed into images or images for telemedicine.¹⁴

Remote dermatology develops in two ways: remote dynamic image or static image promotion. First, it is necessary to connect the image processing equipment between the patient and the remote doctor.¹⁵ This technology is regarded as an auxiliary content of dermatology, but it is very expensive in terms of equipment cost and operating cost. Image mobile consultation usually takes longer than regular consultation. Some experts believe that storing and forwarding static images may be an alternative to dynamic images, rather than consulting, and experts can quickly view the transmitted static images.¹⁶

Telemedicine provides a new way to provide medical services, but accurate and reliable transmission of high-quality images is essential. Especially in dermatology, visual information about the actual condition of the patient is very important. Therefore, real-time telemedicine applications also need these conventional content standards, such as the minimum requirements for graphic analysis, contrast, and color discrimination.^{17,18}

2.4 | Multimedia communication and real-time voice transmission

In the real-time voice system, we implemented multipoint communication so that consultants can communicate and collaborate adequately. This means that if one party speaks, all other purposes of the consulting team can receive the information. In order to achieve this goal, and considering that the signal quality is as good as possible at lower bandwidths, we have adopted a multidistribution technique. In this structure, the distribution program is actually a virtual IP address space used to transmit data to any IP address associated with this distribution group.¹⁹ Each consulting terminal sends the received voice point-to-point to the multidistribution transponder, and the multidistribution transponder sends data to the group terminal through the multidistribution. The system structure diagram is shown as in Figure 1.

Multimedia communication is a key technology in the remote consulting system, and it is also a product of the combination of multimedia technology and network communication technology. Real-time and synchronization are the key features of multimedia communication, as well as the key issue of multimedia communication. Real-time nature requires a large amount of video, audio, animation, and other information to be transmitted quickly to certain network devices.²⁰ Synchronization requires that multimedia information be kept within a time limit. Multimedia communication is one of the key problems to be solved in telemedicine. It is not only used for the transmission of electronic medical multimedia data files, but also for the basic technology of real-time audio and video transmission. This article involves protocol standards related to multimedia communications. On this basis, a real-time voice chat tool based on multipoint communication is realized, and multidistribution technology is used to realize a real multipoint voice subsystem.²¹

2.5 | Remote consultation and remote diagnosis

Remote consultation is an interactive discussion between experts involved in the consultation about the patient's medical images and preliminary diagnosis results. The purpose of remote

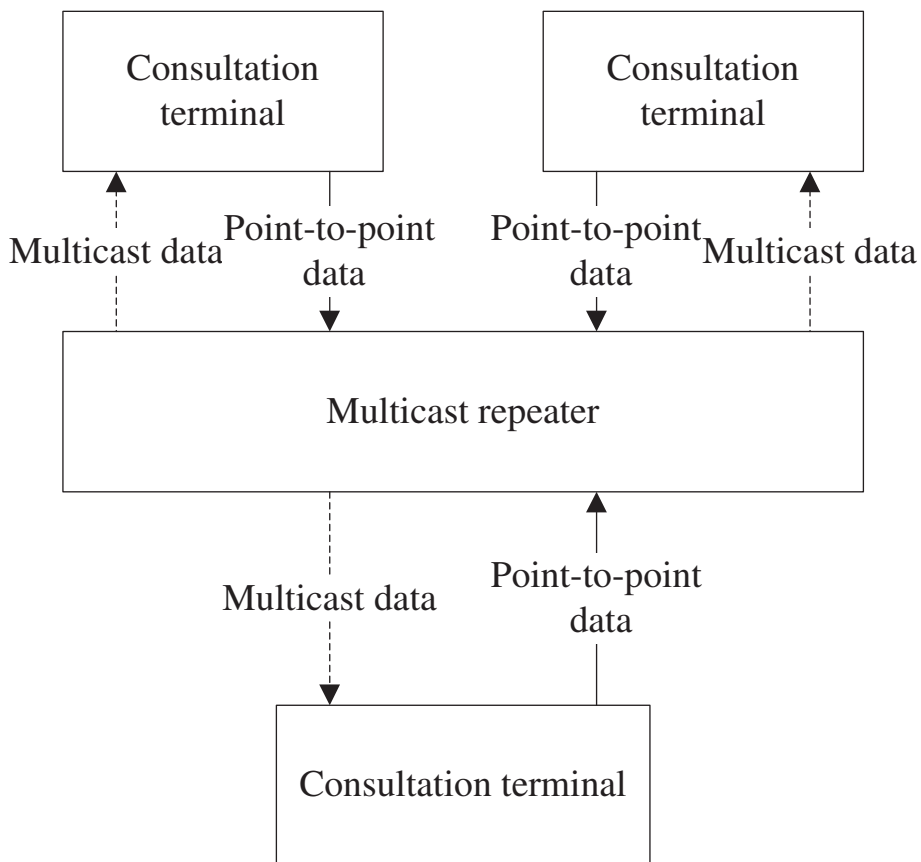


FIGURE 1 Structure diagram of speech system

consultation is to provide referrals for remote doctors and to assist remote doctors in obtaining correct diagnosis results. In this process, in order to support face-to-face conversations between experts, a two-way video conference system is needed to synchronize audio and video signals.²² Currently, remote consulting experts can send images and other files to the remote control while watching and talking, and can send text messages, information, and images through the dashboard. Remote diagnosis is a diagnosis made by doctors by analyzing images and other information of distant patients.

Telemedicine and traditional medical care are complementary and mutually beneficial. The children with skin diseases selected in this study were all admitted to the hospital during the acute exacerbation, and entered the stable period after receiving systemic treatment, and were transferred to the family ward. The family ward can reduce the patient's medical expenses and save the medical resources of medical institutions. In the process of receiving telemedicine technical support in the family ward, if the patient's condition changes, the patient may receive remote consultation for the first time and adjust the treatment plan. If the situation worsens, you can contact the hospital for an appointment as soon as possible. How to choose the right medical method for the right patient is a problem that both telemedicine and traditional medical care should consider in the future.²³



2.6 | ZigBee network routing algorithm analysis and research

As the core content of the ZigBee network layer, the routing algorithm is mainly responsible for the ZigBee network data transmission method, route discovery, and route maintenance. ZigBee mainly supports two routing algorithms: clustering tree algorithm and AODVjr algorithm. The first type is mainly used for tree topology, and the second type is more suitable for grid topology. The ZigBee cluster tree routing algorithm uses a distributed address allocation method to assign unique network addresses to network access nodes.²⁴

The cluster tree algorithm is a relatively simple algorithm that selects the path through the network address based on the parent-child connection relationship. The algorithm does not require additional storage resources and can also store general emission control packages. The network address of a node can be calculated based on the addresses of the parent node and the child node. According to the network depth and network parameters of node d , when a new node wants to enter the network, the router node uses formula (1) to calculate the offset between the addresses and has been assigned to the child nodes:

$$T_{\text{skip}}(e) = \begin{cases} 1 + T_m \cdot (L_m - e - 1) & S_m = 1 \\ \frac{1 + T_m - S_m - T_m S_m^{L_m - e - 1}}{1 - S_m} & \text{other} \end{cases} \quad (1)$$

When a new node joins the node F whose router network address is F_A and becomes its n th child router (terminal) node, the parent node F assigns address T_A to this child node and can be calculated by formula (2).

$$T_A = \begin{cases} F_A + (n - 1) \times T_{\text{skip}}(e) + 1 & T \text{ is the router} \\ F_A + S_m \times T_{\text{skip}}(e) + n & T \text{ is terminal equipment} \end{cases} \quad (2)$$

According to the cluster-tree algorithm, when sending a data packet, the source node uses the network address of the target node to calculate the next hop of the path. For a routing node with direction R and depth e , if the target node is its descendant node, the direction of K will be determined according to the following formula.

$$R < K < R + T_{\text{skip}}(d - 1). \quad (3)$$

If the target node is its affiliated node, the node will send the data packet to its affiliated node. If not, according to formula (4), the data packet is forwarded to its parent node along the hierarchical path.

$$P_R = R + 1 + \left\lceil \frac{K - R - 1}{T_{\text{skip}}(e)} \right\rceil T_{\text{skip}}(e). \quad (4)$$

In order to reduce the overrun caused by the traditional cluster-tree algorithm, the neighbor table is used when forwarding data according to the parent-child connection relationship. If the neighboring node can provide better routing, the neighboring node is selected as the next hop node. For any two nodes, the MA node is defined as the ancestor node with the highest depth



value. According to the neighbor table and the address allocation mechanism, the depth of two nodes can be uniquely determined by their addresses. If a node on the network receives a data packet to be sent to the target node, it can first calculate the number of RC hop routes that will pass the cluster tree algorithm.

$$RC_{TR} = d_m - d_{md} + d_m - d_{md}. \quad (5)$$

If the node N_m has a nonparent nonchild neighbor node N_n with a depth of d_n , the neighbor table can be used to calculate the number of route hops through N_n to N_d :

$$RC_{ETR} = 1 + d_n - d_{nd} + d_n - d_{nd}. \quad (6)$$

The number of hops that can be reduced by adjacent nodes is:

$$\Delta RC = d_m - d_n + 2(d_{nd} - d_{md}) - 1. \quad (7)$$

The principal component analysis method can reduce the dimensionality of multivariate compound factors and combine them into various principal component variables, so as to obtain scientific and effective data information. Assuming that the variable P is observed based on the sample data, the data matrix of N samples is:

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix} = (x_1, x_2, \dots, x_p). \quad (8)$$

Randomly select the selected variable F to P new variable, the main type of function analysis method can be expressed as:

$$F_k = \alpha_{k1}x_1 + \alpha_{k2}x_2 + \dots + \alpha_{kp}x_p, j = 1, 2, \dots, p. \quad (9)$$

The main steps of principal component analysis are as follows:

1. Data standardization processing. It is necessary to eliminate the influence of dimensions between factors and carry out a data standardization process to obtain a standard matrix sample, namely:

$$x_{ik}^* = \frac{x_{ik} - \bar{x}_k}{\sqrt{\text{var}(x_k)}} \quad (10)$$

among them:

$$\bar{x}_k = \frac{1}{n} \sum_{i=1}^n x_{ik}, \quad (11)$$

$$\text{var}(x_k) = \frac{1}{n-1} \sum_{i=1}^n (x_{ik} - \bar{x}_k)^2. \quad (12)$$



2. Calculate the correlation coefficient matrix of the sample:

$$T = \begin{pmatrix} t_{11} & t_{12} & \dots & t_{1p} \\ t_{21} & t_{22} & \dots & t_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ t_{p1} & t_{p2} & \dots & t_{pp} \end{pmatrix}. \quad (13)$$

According to the modeling requirements, assuming that the original data is still used after the normalized data is processed, the correlation coefficient of the normalized data is:

$$t = \frac{1}{n-1} \sum_{i=1}^n x_{ri} x_{rk}. \quad (14)$$

3. Extract the principal components.
4. Determine the number of principal components. Principal component contribution rate is the ratio of the variance of each principal component to the sum of the total variance of all principal components. Assuming the contribution rate is q , the formula is as follows:

$$f = \frac{\gamma_i}{\sum_{i=1}^p \gamma_i}. \quad (15)$$

Multiple linear regression describes the linear relationship between the initial data of multiple independent variables and the data of a single dependent variable. The construction of the model requires iterative calculations based on the initial values of more independent variables and dependent variables to achieve the optimal solution, various regression coefficients and equations, and finally test the equations to ensure validity. The formula for multiple linear regression is as follows:

$$y_i = \beta_i + \mu_{i1}x_1 + \mu_{i2}x_2 + \dots + \mu_{in}x_n + \eta_i. \quad (16)$$

3 | TELEMEDICINE TECHNOLOGY SYSTEM DESIGN TEST

3.1 | Test subject

This article first analyzes the needs of the telematics consulting system. Based on the analysis of remote consultation functions, process analysis and data status, the system is designed in five aspects according to the business process of the system: registration appointment, inspection, consultation, results, and feedback links. The system function requirements in this article are divided into medical record management functions, consultation management functions, and system management functions. While realizing the functions required for traditional telemedicine consultation, the system also models various disease diagnosis, and uses disease templates to assist primary doctors in creating electronic patient records to improve the effectiveness of telemedicine diagnosis and treatment.



3.2 | Test design

The remote consultation system is mainly based on the application of network technology and medical technology. The resources of each hospital can be used interchangeably, so that patients do not have to consider regional differences when treating patients. My country's communication technology and medical technology are developing rapidly and the scale is getting bigger and bigger. The remote consulting system is not only a product of the rapid development of telemedicine technology, but also a key part of constructing a telemedicine system. It is an important platform for patients to communicate with medical experts through all partner hospitals, speeds up the remote exchange of information, and improves the basic technology of remote consultation.

1. Demand analysis. System requirements analysis is the first step in system design. It first analyzes the business process and data process of the system, and then analyzes the operation analysis of the system with the help of case diagrams. This chapter introduces system function requirements, which are mainly divided into medical record information management function, consulting information management function, and system decision management function.
2. System design. The telemedicine medical system uses the B/S model to make the best work plan according to the different environments in which it operates. The system design is optimized for scalability and openness to create a good and complete working environment for the large-scale construction and operation of the hospital.
3. System implementation. Since the overall design of the system is the realization of modules, three basic functional modules of the system are realized: case management, consultation management, and system management functions, including the design of program flowcharts, key codes, and implementation interfaces.
4. System test. With the help of some test tools, start testing related software functions according to the drawings and design steps, and use the corresponding test tools according to the actual situation to protect the system that performs the test operation and check possible error analysis and judgment.

3.3 | Experimental results

The telemedicine system is an important emerging thing in the current telemedicine technology, it can complete an important part of the creation of telemedicine. In addition, patients can undergo video treatments with hospital expertise in any local hospital and require relevant information to ensure rapid and effective creation of medical equipment, so that large hospitals can exchange data and information, and can also conduct continuous medical treatment. The development of operation and medical video technology equipment has greatly promoted the rapid development of medical care. The overall structure design is shown in Figure 2.

Using existing data and information, you can communicate face-to-face with remote experts to complete the quality and timeliness of the diagnosis, which can maximize the efficiency of treatment and medical diagnosis, and greatly ensure health and development. Therefore, the continuous construction and development of information technology and the increasing application and development of network technologies can change the quality and meaning of human medical services to the greatest extent, and can also enhance the core competitiveness of human medical services. As people's income levels increase, people's awareness of self-care also increases, which

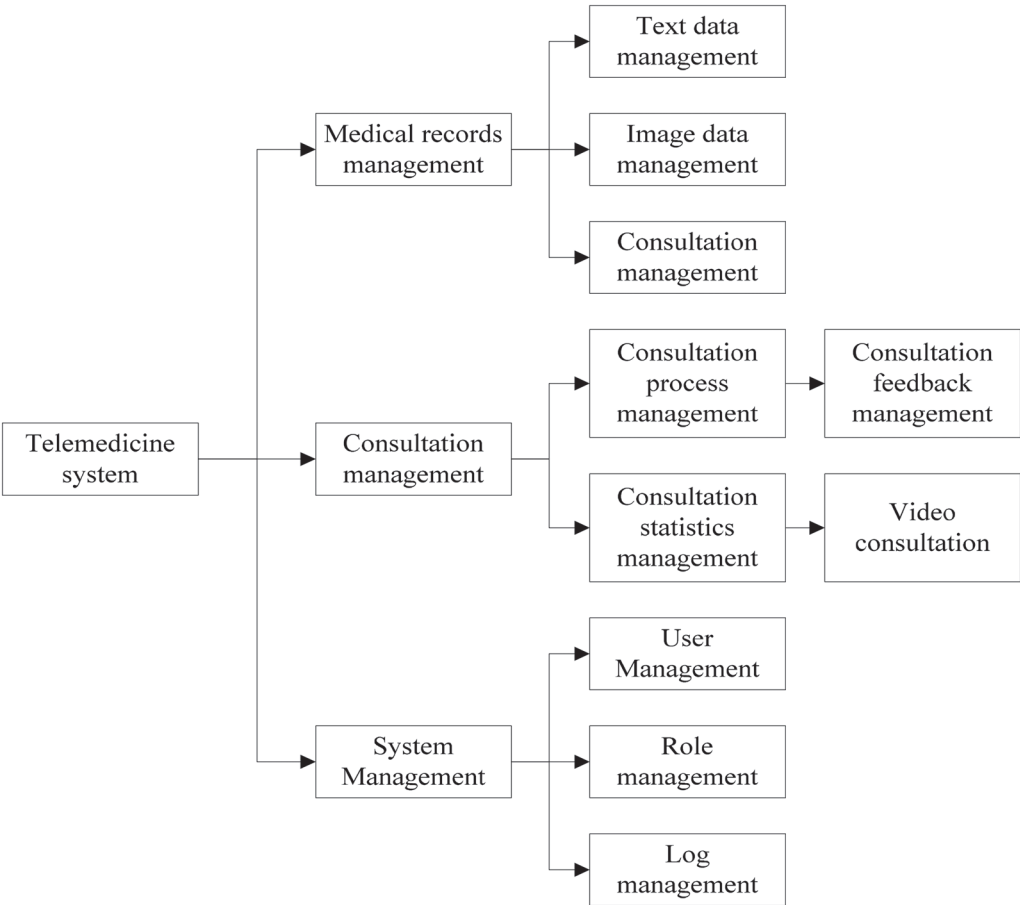


FIGURE 2 Overall system architecture design

also increases the demand for medical resources. Therefore, telematics consultation is also the direction of my country's next medical reform.

4 | TELEMEDICINE TECHNOLOGY BASED ON SKIN IMAGING

4.1 | Effect of telemedicine

Weekly telemedicine videos provide face-to-face medical consultation, which can fully mobilize patients' treatment initiative and enthusiasm, stimulate patients' belief and self-management ability, and shape behaviors in subtle ways. This in itself is also a manifestation of the change in recovery behavior. Once all the rehabilitation training in this study is completed, most patients in the intervention group can continue rehabilitation training and obtain the long-term benefits of rehabilitation training. Guardians are the most important backing for children with skin diseases, and they also play a role in supervising the rehabilitation education of children with skin diseases. Once family members have participated in relevant knowledge training,

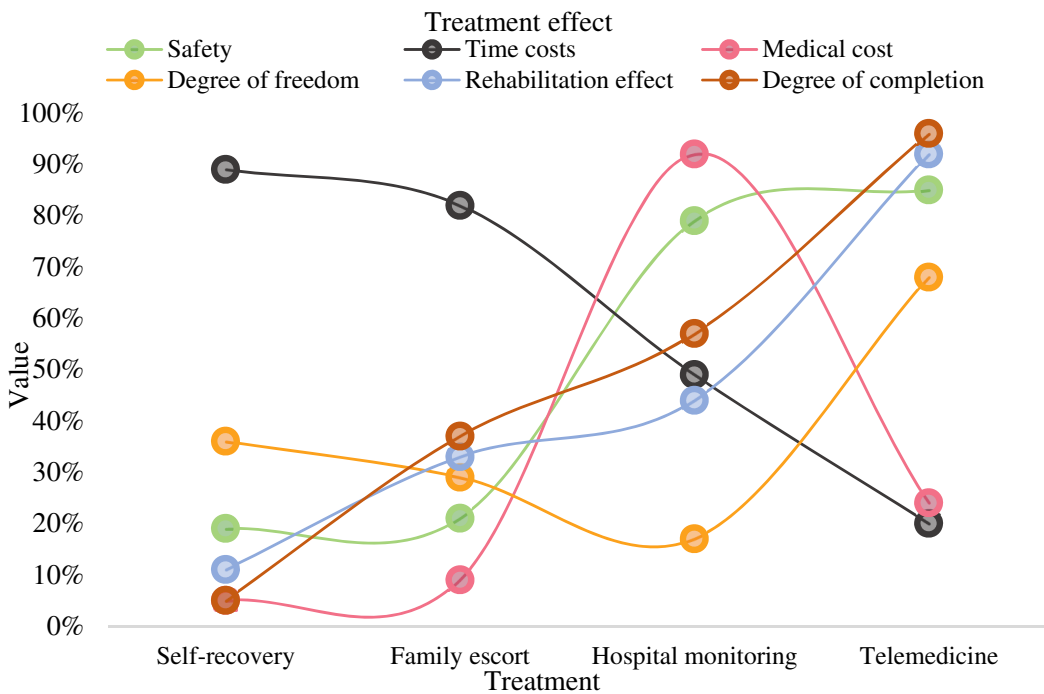


FIGURE 3 Comparison of treatment effects

TABLE 1 Comparison of treatment effects

Treatment	Safety	Time costs	Medical cost	Degree of freedom	Rehabilitation effect	Degree of completion
Self-recovery	19%	89%	5%	36%	11%	5%
Family escort	21%	82%	9%	29%	33%	37%
Hospital monitoring	79%	49%	92%	17%	44%	57%
Telemedicine	85%	20%	24%	68%	92%	96%

a preliminary understanding of the rehabilitation of patients with skin diseases and a further understanding of telemedicine technology will promote the cooperation between physicians and patients. In the self-recovery method, although the cost of medical care only accounts for 5%, the cost of time accounts for 89%, and the recovery effect and safety are as low as 11% and 19%. Using telemedicine, the safety, rehabilitation effect and completion rate are as high as 85%, 92%, 96%, while saving a lot of time and medical costs. As shown in Figure 3 and Table 1.

4.2 | Evaluation results of the telemedicine use behavior model

In the entire weighting table, the weights of the six observation variables of physicians' persistence and confidence in telemedicine use are relatively low, which indicates that these observation variables have relatively weak explanatory power for indirect variables of

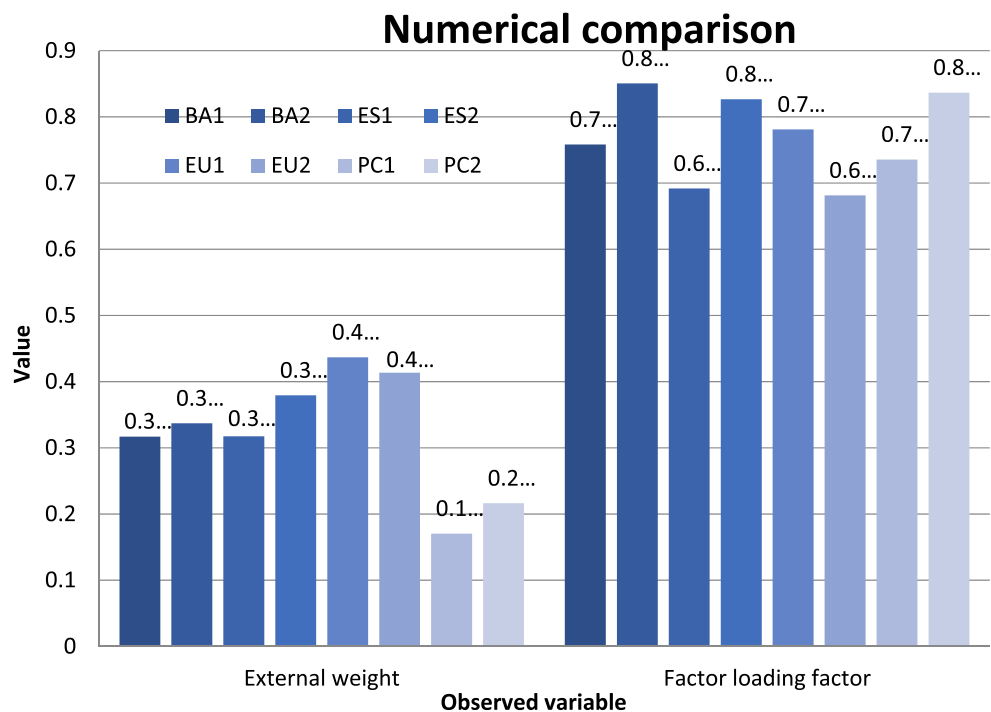


FIGURE 4 External model weights and factor loading coefficients

persistence and confidence. In the entire weighting table, doctors believe that the weights of the seven observation variables of telemedicine utility are relatively low, which indicates that the ability of these observation variables to explain the indirect variables of perceived utility is relatively weak. The table shows that most of the observed variables have relatively high load factors. Among the 12 load factors, two load factors are between 0.6 and 0.7, five are between 0.7 and 0.8, and five are between 0.8 and 0.9. Load factors are above 0.9, 0, indicating that the observed variables reflect the height of their respective latent variables. As shown in Figure 4 and Table 2.

4.3 | Key technologies

Telemedicine focuses on the communication of medical information, such as various medical images, patient records, and medical information. The DICOM standard is an international de facto standard for medical digital imaging and communication. Therefore, in the process of integrating hospital information systems with video conferencing software, it is necessary to develop appropriate tools to fully understand the quality of medical information. Communication technology, computer technology, and image processing technology have made considerable progress. Various international organizations and teams have developed a series of high-performance multimedia algorithms and standards, and network technology has rapidly become popular. Telemedicine has gained a specific area of technological development, and remote diagnosis has become more feasible. Based on this background, this article focuses on system architecture, multimedia communication, real-time voice transmission, WEB-based electronic medical records,



Observed variable	External weight	Factor loading factor
BA1	0.316825	0.758213
BA1	0.336971	0.850317
ES1	0.317365	0.691743
ES2	0.379236	0.826643
EU1	0.436791	0.781041
EU2	0.413596	0.681371
PC1	0.170326	0.735519
PC2	0.216315	0.836594
UF1	0.192341	0.756721
UF2	0.196028	0.796492
VL1	0.364987	0.807762
VL2	0.362615	0.801202

TABLE 2 External model weights and factor loading coefficients

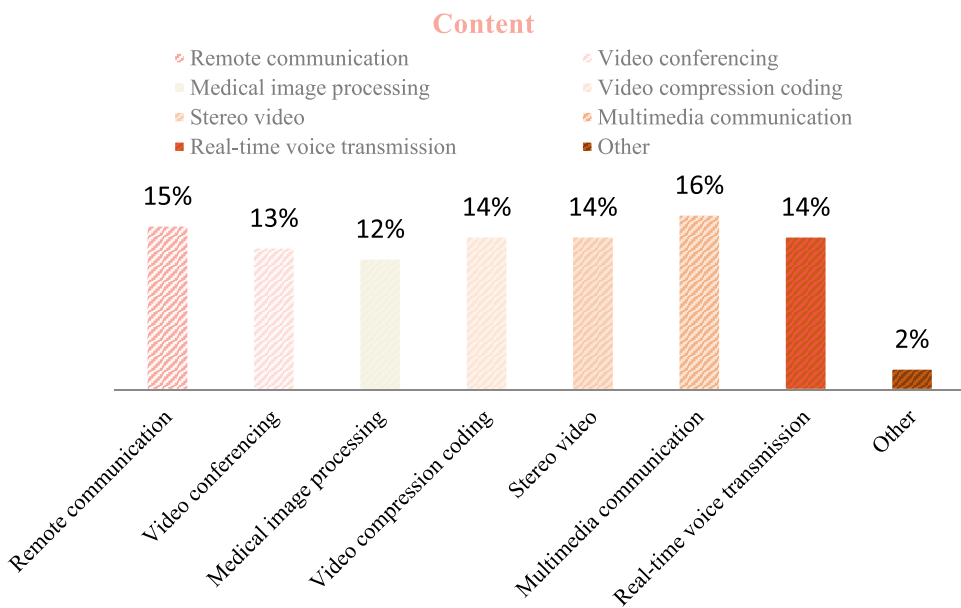


FIGURE 5 Related technologies involved in the system

collective diagnosis, and other key technologies related to the design and implementation of telemedicine systems. As shown in Figure 5 and Table 3.

4.4 | Distribution of telemedicine consultation departments

According to the statistics of the Telemedicine Center of Province A, from January 1, 2016 to December 31, 2019, a total of 150 municipal and county-level hospitals in the province applied

TABLE 3 Related technologies involved in the system

Related technology	Content
Remote communication	The telemedicine system can adopt a variety of communication network technologies, corresponding to different communication networks and actual applications, and can adopt a variety of application modes.
Video conferencing	A comprehensive multimedia remote interactive platform integrating audio-visual interaction, text communication, image processing, document sharing, and file transmission is realized.
Medical image processing	The process of converting an image signal into a digital signal and processing it with a computer. Including image coding, image segmentation, image recognition, and so forth
Video compression coding	H.264 is a new video codec standard proposed by JVT to achieve higher video compression ratio, better image quality, and good network adaptability.
Stereo video	The addition of scene depth information can enhance the realism and fidelity of visual effects, and it is also more conducive to improving the diagnostic efficiency and accuracy of medical experts.
Multimedia communication	It is necessary to transmit a large amount of real-time voice and video information. When transmitting between multiple points, it is necessary to ensure the real-time performance of the transceiver.
Real-time voice transmission	In order to enable consultation experts to achieve sufficient communication and collaboration, multipoint communication is realized, that is, when one end speaks, all other ends in the consultation group can receive information.
Other	This article also involves key technologies such as system architecture, WEB-based electronic medical records, and collaborative diagnosis.

for consultation with the telemedicine center. The number of applications for remote consultation services of cooperative hospitals is increasing every year. The average annual number of consultation services is 14,926, and the average annual growth rate is 110.62%. A total of 603,494 remote consultations have been conducted, which shows that the demand for remote consultation in the city and county cooperative hospitals is increasing. According to the teleconsultation status of the telemedicine center, there are 59 clinical science and medical technology departments for remote consultations belonging to eight departments, and the main departments for remote consultations are internal medicine, surgery, and women and children's departments. The total can reach 74.87%, internal medicine accounts for 40.34%, surgery accounts for 22.86%, and gynecology and children accounts for 11.67%. In the case of remote consultation, the professional fields and the top 10 in total consultation are: Neurology, Respiratory Medicine, Pediatrics, Orthopedics, Oncology, Cardiovascular Medicine, Neurosurgery, Hepatobiliary and Pancreatic and Liver Transplantation, Gastroenterology, Gynecology. As shown in Figure 6 and Table 4.

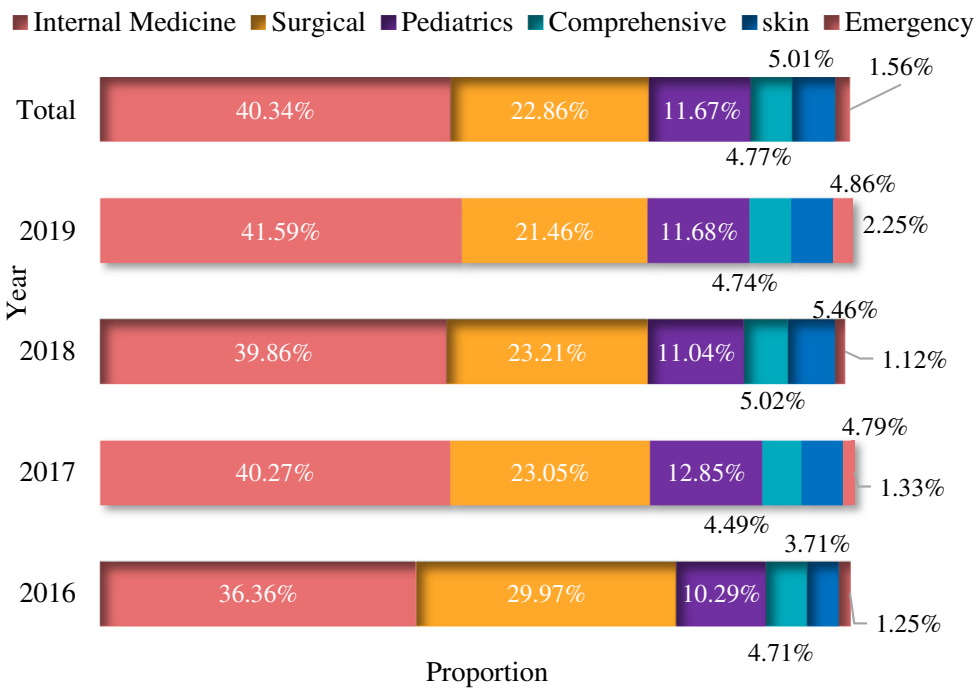


FIGURE 6 Clinical cases of remote consultation

TABLE 4 Clinical cases of remote consultation

Classification	Internal medicine	Surgical	Pediatrics	Comprehensive	Skin	Emergency
2016	36.36%	29.97%	10.29%	4.71%	3.71%	1.25%
2017	40.27%	23.05%	12.85%	4.49%	4.79%	1.33%
2018	39.86%	23.21%	11.04%	5.02%	5.46%	1.12%
2019	41.59%	21.46%	11.68%	4.74%	4.86%	2.25%
Total	40.34%	22.86%	11.67%	4.77%	5.01%	1.56%

4.5 | Multifactor logistics regression analysis with remote service volume

Through the logistics regression analysis of multiple categorical variables, the factors affecting the amount of remote consultation services were determined. Taking the number of remote prompts as an independent variable, the number of consultations is divided into three levels: 51–100 for level 0 to 1, 51–100 for the second level, and 100 or more for the third level. Nine factors include hospital level, whether medical insurance reimbursement, whether the consultation administrator is full-time, the degree of importance of the department doctor, the economic reward of the department doctor's consultation, the patient's evaluation of the consultation cost, the technical level of the expert, the problem-solving ability of the expert, and the smooth communication between the consultation expert and the doctor. As the dependent variable, statistical analysis

TABLE 5 Multifactor logistics regression analysis

Variable	β	SE	Wald	P	RR	95% CL
Reimbursement	4.026	1.492	9.223	0.005	0.013	0.182
Full-time staff	2.988	1.623	8.177	0.003	0.062	0.301
Economic reward	4.371	2.051	9.151	0.002	0.023	0.152
High technology	3.255	1.577	4.926	0.047	0.031	0.447
Value	2.668	1.506	6.882	0.038	0.049	0.226
Publicity	3.003	1.464	6.255	0.018	0.041	0.351
Solving ability	2.232	1.354	5.941	0.030	0.025	0.214
Smooth communication	1.396	1.227	7.961	0.013	0.113	0.264

was performed using SPSS. Check the important factors of the analysis results $P \leq 0.05$, as shown in Figure 7 and Table 5.

4.6 | Telemedicine service cost indicators

This article uses the specific situation of Province A as a starting point to identify the data, studies various limiting factors in detail, combines relevant telemedicine statistics, and

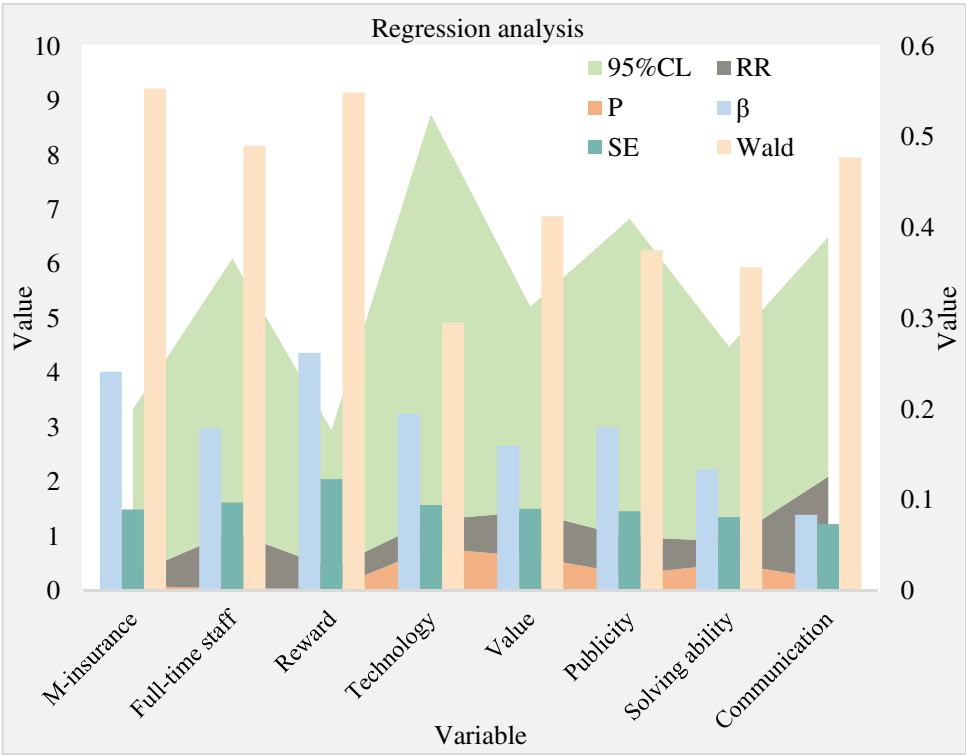


FIGURE 7 Multifactor logistics regression analysis



Cost index

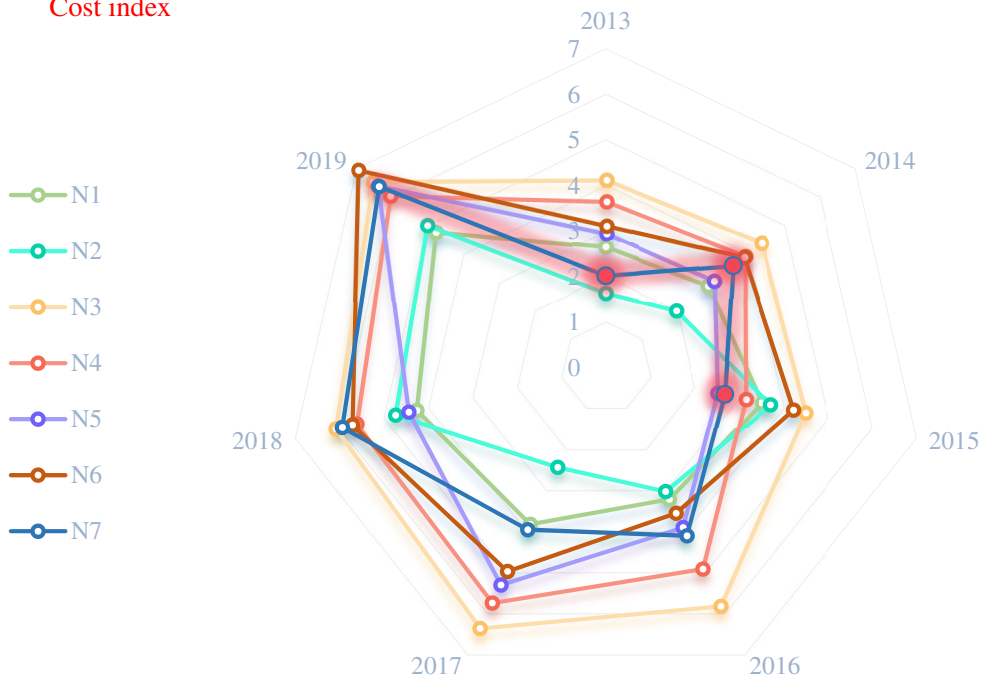


FIGURE 8 Telemedicine service cost data

TABLE 6 Telemedicine service cost data

Year	N1	N2	N3	N4	N5	N6	N7
2013	2.64	1.63	4.11	3.64	2.94	3.11	2.01
2014	2.88	1.99	4.37	3.91	3.05	3.88	3.58
2015	3.51	3.71	4.51	3.15	2.52	4.23	2.68
2016	3.22	3.03	5.83	4.92	3.91	3.56	4.11
2017	3.83	2.44	6.37	5.75	5.31	4.98	3.96
2018	4.26	4.74	6.08	5.61	4.44	5.71	5.94
2019	4.77	5.01	6.53	6.04	6.39	6.94	6.37

selects relevant data between 2012 and 2019 as historical data. For those with missing years, some data are scientifically supplemented by corresponding methods. Related policies, GDP per capita, medical expenses per patient per hospital, the number of tertiary hospitals, and the number of general practitioners per 10,000 population belong to the official website data. The specific data of software and hardware equipment investment, maintenance and depreciation, subcenter construction, medical technician funding, and various remote consultations are derived from the cost data record table of the remote medical center of A province, the historical data record of the remote medical service system platform of the province A, data Honest, and reliable. The specific data is shown in Figure 8 and Table 6.



5 | CONCLUSION

Since the beginning of this century, with the extensive establishment of tele-information centers in well-known domestic hospitals and medical colleges and the development of various telemedicine virtual reality technology work, my country's telemedicine virtual reality technology research has entered a peak period. At present, telemedicine virtual reality technology has been practically applied in many hospitals across the country, and broadcast live at various large-scale international patient diagnosis conferences, providing convenience for many difficult and severe patients and achieving effective treatment results. At present, my country's telemedicine virtual reality technology has been recognized and supported by many researchers at home and abroad, both in academic achievements and practical applications.

In recent years, with the development of society and the improvement of living standards, people have put forward higher requirements for medical services, which has led to the development of telemedicine systems based on medical information technology and network technology. It has changed the traditional medical model, making it the largest growth space and growth potential. Hospitals and hospitals, hospitals and patients directly cooperate through telemedicine virtual reality technology to establish new relationships. Patients and lower-level hospitals can easily enjoy the rich medical resources of the comprehensive central hospital and complete remote consultation, remote education, remote monitoring, and other projects to achieve the purpose of improving medical effects. With the advancement of technology, telemedicine will be further improved and developed.

Our country has a vast territory, abundant resources, and a large population. The distribution of medical resources is extremely uneven. High-quality medical resources are mainly distributed in large and medium-sized cities, and the medical level of grassroots and remote areas needs to be improved. With the improvement of people's living standards, the medical team's requirements for medical service levels are getting higher and higher. Minor illnesses go directly to large hospitals for treatment. This has led to the accumulation of large hospitals and the continuous emergence of primary hospitals, resulting in scarcity and scarcity of high-quality medical resources. The problem of waste, difficult and expensive medical visits is further exacerbated. With the further reform of the medical system, relying on the rapid development of information technology, telemedicine has become the basic idea to solve the problem of difficult and expensive medical treatment.

ORCID

Fei Tian  <https://orcid.org/0000-0001-9764-2795>

REFERENCES

1. Stotts MJ, Grischkan JA, Khungar V. Improving cirrhosis care: the potential for telemedicine and mobile health technologies. *World J Gastroenterol*. 2019;25(29):3849-3856.
2. Kunikowska J, Zemczak A, Górska M, et al. TeleNEN as a telemedicine model for neuroendocrine neoplasm management in case of Meckel's diverticulum NET. *Endokrynol Pol*. 2018;69(3):313-317.
3. Zack B, Arrandale VH, Holness DL. Preventing occupational skin disease: a review of training programs. *Dermatitis*. 2017;28(3):169-182.
4. Elsner P, Bauer A, Diepgen TL, et al. Position paper: telemedicine in occupational dermatology – current status and perspectives. *J Dtsch Dermatol Ges*. 2018;16(8):969-974.
5. Jang Y, Kyun KH. An analysis on the image color of skin expression makeup on color measured values. *J Korea Soc Color Studies*. 2017;31(2):63-73.

6. Delyon J, Ortonne N, Sbidian E, Chosidow O. Keratinocyte dystrophy as a marker of low-dose methotrexate-induced skin toxicity: comment on the clinical image by Mnch et al. *Arthritis Rheumatol*. 2016;68(7):1790-1791.
7. Hatef E, Alexander M, Vanderver B, Fagan P, Albert M. Assessment of annual diabetic eye examination using telemedicine technology among underserved patients in primary care setting. *Middle East Afr J Ophthalmol*. 2017;24(4):207-212.
8. Van Geel MJ, Maatkamp M, Oostveen AM, et al. Comparison of the dermatology life quality index and the children's dermatology life quality index in assessment of quality of life in patients with psoriasis aged 16–17 years. *Br J Dermatol*. 2016;174(1):152-157.
9. Skripov VS, Semenova NV, Kochorova LV, et al. Telemedicine technologies in psychiatry and narcology: specialists' point of view. *Med Technol Assess Choice*. 2019;4(38):49-54.
10. Ayatollahi H, Ghalandarabadi M. A review of the use of telemedicine technology in caring for high-risk pregnant women. *Med Technol J*. 2017;1(4):135-135.
11. Latifi F, Alizadeh S. The influence of national factors on transferring and adopting telemedicine technology: perspectives of chief information officers. *Int J E-Health Med Commun*. 2016;7(3):52-65.
12. Jacob SE, Hill H, Lucero H, Nedorost S. Benzoate allergy in children—from foods to personal hygiene products. *Pediatr Dermatol*. 2016;33(2):213-215.
13. Paul DL. Telemedicine: as the technology has matured, how have the technology management challenges changed? *Int J Electron Healthc*. 2016;9(2-3):186-209.
14. Perrin S. Development and characterization of an optical coherence tomography micro-system: application to dermatology. *Water Res*. 2016;47(19):6992-7005.
15. Cegarra-Sanchez J, Cegarra-Navarro JG, Wensley A, Diaz Manzano J. Overcoming counter-knowledge through telemedicine communication technologies. *J Health Organ Manag*. 2017;31(7–8):730-745.
16. Simpson EL, Bruin-Weller MD, Eckert L, et al. Responder threshold for patient-oriented eczema measure (POEM) and children's dermatology life quality index (CDLQI) in adolescents with atopic dermatitis. *Dermatol Ther*. 2019;9(4):799-805.
17. Huang KTL, Lu TJ, Alizadeh F, Mostaghimi A. Homebound patients' perspectives on technology and telemedicine: a qualitative analysis. *Home Health Care Serv Q*. 2016;35(3–4):172-181.
18. Doarn CR, Merrell RC. A novel approach to telemedicine video control technology (Retraction of vol 15, pg 859, 2009). *Telemed e-Health*. 2016;22(7):622-622.
19. Deborah S. New computing technology ensures safety for telemedicine and Telemonitoring in oncology care. *ONS Connect*. 2016;31(12):32-33.
20. Bonner J. Technology advances: harnessing the telemedicine beast. *BSAVA Companion*. 2017;2017(6):12-14.
21. Ardalan K, Switzer GE, Zigler CK, Hershey NM, Torok KS. Psychometric properties of the children's dermatology life quality index in pediatric localized scleroderma. *J Scleroderma Relat Disord*. 2018;3(2):175-181.
22. Gnana V, Taneja N, Nirupama AY. Telemedicine: new technology, new promises? *Indian J Community Health*. 2019;31(4):437-441.
23. Kathleen M, Leafman JS, Horton MB. Access to digital communication technology and perceptions of telemedicine for patient education among American Indian patients with diabetes. *J Health Care Poor Underserved*. 2017;28(4):1522-1536.
24. Akakpo AS, Teclessou JN, Saka B, et al. Profile of erythroderma in children in hospital dermatology departments in Lomé, Togo: study of 28 cases. *Med Sante Trop*. 2018;28(3):277-279.

How to cite this article: Gao J, Lyu C, Qiao X, Tian F. Telemedicine virtual reality based skin image in children's dermatology medical system. *Computational Intelligence*. 2021;1–20. <https://doi.org/10.1111/coin.12458>